

A meta-analysis of iron-based additives on enhancements of biogas yields during anaerobic digestion of organic wastes

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ABSTRACT

The profitability of biogas industries largely depends on substrate biodegradability. Most of the substrates for sustainable energy recovery via anaerobic digestion like lignocellulosic materials contains complex polymeric substances and requires some enhancements to achieve optimum biodegradability. Previous studies have reported that the use of iron additives improved both degradability of such substrates and biogas yield, but iron-based enhancements have shown variability in the input and yield results. This evidence-based study used meta-analysis in determining the performances of iron-based additives on biogas yield. Previously published articles on the subject were systematically identified and compiled based on set inclusion criteria. The database was used to analyze the influence of iron-based additives on the enhancement of biogas yield during anaerobic digestion of organic wastes. The pooled effect estimates at 95% confidence interval was generated with the use of Random-effects model. The overall result of the meta-analysis performed showed that iron-based additives supplementation significantly enhanced biogas yield at Standard mean difference = 1.451, 95% CI: 1.280–1.622, $p < 0.001$. Restricted subgroup analysis indicated that all the moderators had a significant influence on biogas yield of iron-based treated anaerobic digestion processes at $p < 0.001$. Evidence of heterogeneity was observed, while meta-regression analysis indicated that most of the sources of heterogeneity were explained by the additive-types, substrate-types, dosage range and research location at $p < 0.05$. The result of this study showed the potential of iron-based additives supplementation to improve biogas yield during anaerobic digestion of organic wastes. The information from this study avails important suggestions on policy advancements and sustainable usage of iron-based enhanced anaerobic digestion processes in biogas industries.

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1. Introduction

Bioconversion of biomass to biofuel is a sustainable means of reducing dependence on fossil energy sources in the face of increasing energy demands, ensuring eco-friendly resource recovery and appropriate waste management strategies (Hiloidhari et al., 2014; Gao et al., 2018). The higher percentage of biomass composition for anaerobic digestion are natural polymers (lignin, cellulose, hemicellulose, etc.) with structures that take longer time to breakdown and unlike monomers, are not easily biodegradable (Akunna, 2018; Li et al., 2018). Aside the issues of lignocellulosic structural degradation, anaerobic digestion is limited by slow

syntrophic metabolism of intermediate products and low acidification rate (Zhang et al., 2019; Bharathiraja et al., 2018). Difficulties in removing humic substances and bacterial cell disintegration also inhibits methanogenic activities during the digestion process (Liu et al., 2016; Hao et al., 2017). Owing to the above reasons, achievement of higher biogas yield requires substrate pretreatments and enhancements with additives (Li et al., 2018; Ye et al., 2018).

Although pretreatment of substrates facilitates the break-down of its lignocellulosic content, inhibition of methanogenic activities arising from extreme pH values, high energy input and increased amount of salt formations are part of their disadvantages (Zhang et al., 2014). However, previous studies have reported that enhancement of anaerobic digestion with additives reduces toxins, increases the speed of startup, supplies deficient nutrients, reduces

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hydraulic retention time (HRT), increases digester stability and improved rate of biogas yield increased (Ye et al., 2018; Kim et al., 2017; Baredar et al., 2016). Amongst other enhancement options with trace elements, the ability of iron to lose or gain electrons, positions it as ideal and versatile in enhancing methanogenic activities at appropriate dosages. Addition of iron salts during anaerobic digestion resulting in iron homeostasis is essential to anaerobes (bacteria colony) enhancement, DNA replication, increased methane yield and enhancement of electron transfers (Zhang et al., 2019; Casals et al., 2014).

Previous investigations showed that various synthetic and natural iron-based additives had either enhanced or inhibited anaerobic digestion in both laboratories and industries (Casal et al., 2014; Wang et al., 2018). Zhang et al. (2019) and Cruz Viggí et al. (2014) reported that the use of magnetite increased, methane yield by 16.1% and aided propionate conversion rate and enhanced direct interspecies electron transfer (DIET). Casals et al. (2014) reported that addition of Fe₃O₄ NPs to an anaerobic process achieved 180% rise. The use of WISs during anaerobic digestion resulted in increased methane production of 21.4% (Hao et al., 2017). Similar successful enhancements with NZVI recorded with 88% rise in methane yield (Amen et al., 2017). Amen et al. (2018) reported that lower doses of NZVI inhibited biogas yield by 16.2%, while Wang et al. (2018) reported higher yield with lower doses of NZVI and inhibition of 54.7% was detected at higher doses of 10 mg/L. Yang et al. (2013) reported inhibition in all the doses of NZVI studied. Although most literature reported higher yield with the use of trace iron-additives, Lui et al. (2019) also reported inhibition with the addition of trace ferroferric oxide. Aside the inhibition or enhancement of biogas yields arising due to the disproportionate addition of iron supplements, the environmental impact of non-biodegradable compounds (vivianite, siderite, etc.) formed during anaerobic digestion is of huge economic implications to biogas industries (Puyol et al., 2018).

These substantial inconsistencies in the reported results of enhanced anaerobic digestion processes with iron-based additives and their adverse impacts, even when the same substrates, additives and dosages were used is a source of concern to biogas firms. According to Strömberg et al. (2014), Holliger et al. (2016) and Abdelsalam et al. (2017), these variabilities emanates from substrate heterogeneity (origin, sizes and compositions), inoculum viability and source, iron-additives types (nanoparticle or trace, sizes and chemical compositions), dosages, operational conditions, etc. Although some international multi-laboratory efforts, systematic analyses and reviews, aimed at resolving these dissimilarities (Angelidaki et al., 2009; Raposo et al., 2011; Lavergne et al., 2018), no known comprehensive effort at collating the impacts of all previously studied influence of iron-based additives on anaerobic digestion process have been undertaken.

The meta-analysis technique is an advanced statistical means of resolving uncertainties associated with divergence in research results through data aggregation from multiple studies. It combines information and evaluates treatment effect that may have eluded the effect size of individual study (Van Houwelingen et al., 2002). In biogas production, few systematic reviews have been undertaken. These includes understanding how variability in operating conditions can aid standardization of anaerobic digestion process (Lavergne et al., 2018) and analyzing the microbial diversities in biodigesters (Nelson et al., 2011). In the same vein, Miranda et al. (2015) and Miranda et al. (2016) reported on the meta-analysis of assessing the methane yield from dairy manure and possible greenhouse gas emissions anaerobic digestion. Cao et al. (2019) reported on the meta-analysis of the influence of different additives on mitigation of emission of methane and other gases.

However, none have reported on the conduct of meta-analysis study on the effects of iron-based additive usage on biogas yield during anaerobic digestion, hence this research.

In bridging the identified research gap in this systematic study, we quantitatively aggregated and analyzed results of the impacts of iron-based additives on biogas yield during anaerobic digestion of organic wastes from previous studies. This is aimed at availing biogas industries and researchers of needed information on the influences of the iron additives and other explanatory variables (substrate types, dosages and operational conditions) on enhanced biogas yields. This study will involve the creation of the needed database and systematic analysis of resulting information.

2. Materials and methods

2.1. Literature search and study selection

A bibliographic search of iron-based additives and biogas production with the use of databases such as the Web of Science, Google Scholar, Scopus, Science Direct, etc. to assess published referenced and unreferenced studies in the English language were completed in July 2019. With the aid of proximity searching ("..." or "..."), Boolean logic (NOT, AND, OR), alternate spelling (?), wildcard (\$ or *) and integrative list of search words like *iron additives, trace elements, nanoparticles, anaerobic digestion, bio-digestion, biomass, organic wastes, agro-wastes, sludges, dosages, enhancements, inhibitions, biogas yield, biomethane potentials, methane production, mesophilic temperature, effective biodigester volume, biodigester types, methanogens, hydraulic retention time, C: N ratio, and biodegradation*. Possible publication bias occurrence from positive findings was reduced by surveying relevant lists of publication references and searching out "grey publications" from the internet (Sabater et al., 2018). Over 900 publications were obtained and were assessed individually for selection based on set criteria.

2.2. Data extraction

After the literature survey of relevant research outputs on iron-based enhancement of anaerobic digestion published until July 2019 were conducted. Only one hundred (100) published studies out of over 900 selected studies met the set criteria for inclusion in the database for meta-analysis. The studies were included if: (a) measurable data were presented for determination of average value, estimation of sample size and data variability from both random and control trials; (b) availability of information on the type of iron-based additives (liquid or powder, nanoparticle or trace or scrap or ore), substrate types (organic wastes) and dosages; (c) at least an expected outcome and a measure of dispersion (e.g. standard deviation (SD), standard error (SE) and confidence interval (CI) of 95% (note that SE and 95% CI were converted to SD)) was reported (Sabater et al., 2018; Lavergne et al., 2018).

However, rejected papers were largely based on data duplication, lack of randomization, reviewed paper and studies not conducted on the impact of iron-based additives or on organic wastes (not synthetic or chemicals substrates). Data used for this meta-analysis were obtained directly from tables, plots and intext data of each selected publication. More so, information on explanatory variables like author names, countries of research, year of publications, substrate types, C: N ratios, operational variables (digestion period or HRT, digester volume, and temperature) and variability measures (SE, SD and 95% CI) were collected. The standard deviation (SD) is an important measure of dispersion for meta-analysis, but in several studies, SD of yield results was not available, hence they are estimated. SD values are estimated from SE (if provided) using an equation ($SD = SE * \sqrt{n}$), where n = number of

replications (Higgins and Deeks, 2011; Ogbuewu et al., 2019). Similarly, lots of studies neither reported SD nor SE values, in such cases, 10% of mean variables were assigned as the SD values (Luo et al., 2006; Adu et al., 2019). The influence of the estimated SD on the result was considered with sensitivity analyses (Adu et al., 2019). The retrieved data from the qualified primary publications were converted to a CSV file format which is the acceptable file format for the meta-analysis software.

2.3. Statistical analysis

All relevant data extracted were saved in CSV format of Microsoft Excel (2016) for further analyses. OpenMEE meta-analysis software by Brown University (Wallace et al., 2017) was used for meta-analysis, meta-regression and predictor-treatment interaction determination. The study location plot was performed with Tableau version 2019.4. Other analyses and plots were done with Microsoft Excel 2016.

2.3.1. Effect size and meta-regression

The effect size which according to Borenstein et al. (2009) and Sullivan and Feinn (2012) refers to the magnitude of difference between variable means and the extent of treatment effect (in this case, the effect of iron-based additives) was determined in this meta-analysis study. Hedges' d was used and the standard mean difference (SMD) between treatment and control biogas yields was also used to calculate the effect size (Borenstein et al., 2009). Hedges' d which is the most preferred and used metric in meta-analysis, has a major advantage over response ratio because, it is usable for small sample size studies and not influenced by unequal sampling in a paired group (Hedges and Olkin, 1985; Morales and Traveset, 2009; Adu et al., 2018). Hedges' d was estimated using the below-stated equation (1) and the variance of Hedges' d is given in equation (2) (Rosenberg et al., 2013):

$$d = \frac{(\bar{X}_e - \bar{X}_c)}{S} J \quad 1$$

$$V_d = \frac{N_e + N_c}{N_e N_c} + \frac{d^2}{2(N_e + N_c)} \quad 2$$

Where $\bar{X}_e$ and $\bar{X}_c$ are respectively the means of experiment and control groups.

N_e and N_c are the sample size of the experimental and control groups respectively

J is a correction term for a small sample size $(1 - \frac{3}{4(N_e + N_c - 2) - 1})$.

S is the pooled standard deviation $(\sqrt{\frac{(N_e - 1)S_e^2 + (N_c - 1)S_c^2}{N_e + N_c - 2}})$.

The DerSimonian and Laird test (Q -statistic) and inconsistency index (I^2 statistic) were used in determining heterogeneity (Higgins and Thompson, 2002). The designation of I^2 values as low (25% and below), moderate (50%) and high (75% and above) are based on Higgins and Thompson (2002) classification. The grand mean (pooled) effect size of iron-based additives impact on biogas yield was determined with a random-effects model (REM) according to DerSimonian and Laird (1986) methods. Since the data used in the study varies from each other, the between-studies (Q_B) and within-studies variance were ascertained in addition with the sampling error using the restricted maximum likelihood (REML) method and mixed-effects models (Borenstein et al., 2009). The interaction effects between treatment and covariates were also determined (Simmonds and Higgins, 2007; Wallace et al., 2017).

2.3.2. Forest plot

The overall effect of iron-based enhancement on biogas yield was pictorially represented in the form of forest plots (Borenstein et al., 2009; Ogbuewu et al., 2019). The thick middle line is known as the line of no effect, points to its right depict an increase in biogas yield, while points to the left show inhibition or decrease in biogas yield. The lines joining the black squares (effect sizes) in each study are the upper and lower 95% CI for the effect size. The effect size inverse of variance depicts each study weight; larger squares entails large weight (Lean et al., 2009). The dotted line represents the overall effect estimates. When the diamond at the base touches the line of no effect, it shows that the pooled estimates are not significant. According to Cohen's (1988) recommendation, effect sizes were interpreted as small (<0.2), moderate (0.5), and large (>0.8).

2.3.3. Sensitivity analysis and publication bias

The robustness of the meta-analysis result was determined via sensitivity analysis. Sensitivity analyses are often deployed in knowing the impact of studies seen as having unwarranted influence in the meta-analysis (Lean et al., 2009). In this study, it was conducted by leaving-out-one study prior to running the analysis. Extracted data were segregated in sub-groups for analysis and meta-regression according to the following explanatory variables: additive types, country of study, dosages, substrate type, bioreactor effective volume, HRT, temperature and C: N ratio. The proof of publication bias was evaluated with the aid of Rosenberg's fail-safe number (Nfs), this avails the number of studies with non-significant results that would be required to nullify the overall effect size. According to Borenstein et al. (2009), meta-analysis outcome can be referred to as being robust despite the possibility of publication bias when the fail-safe number (Nfs) is greater than $(>5*(n) + 10)$, where n = number of effect size.

3. Results

3.1. Overview of data

After screening for inclusion, 100 studies and three hundred and forty-one studies (341) reporting on were selected for meta-analysis. The Table 1 shows the moderator levels, definitions and percentages of included studies. Seventeen (17) different iron-based additives for enhancement of biogas yield were identified as shown in Table 1. The most commonly used were NZVI (21.4%), Fe_3O_4 NPs (18.8%), ZVI (15.2%) and the least used was $Fe(OH)_3$ (0.3%). For the analysis of substrates, sludge (58.1%) was the most reported substrate used with iron-based additives, food wastes (8.5%), combined organic wastes (8.2%), cattle manure (7.0%) and the least used was green algae (0.6%). Most used substrates had low range of C: N (<20) ratio (55.8%), the C: N range (20–30) which according to EPA (2012) is the optimum range was 18.6% and the high range was 25.7%. Since most reported substrate C: N ranges were out of the recommended range, co-digestion of low and high C:N substrates is most desirable for increased profitability in biogas industries (EPA, 2012; Ugwu and Enweremadu, 2019). The moderate laboratory batch bioreactor with effective volume capacity ranging between 310 and 900 mL (43.1%) was the most used in the included studies, while the semi-continuous reactor-type (both moderate (0.6%) and large-sized reactors (2.3%)) were the least used. Mesophilic temperature (96.4%) was the most reported anaerobic digestion condition in the included studies. The hydraulic retention time (HRT) ranges of 31–60 (35.5%), 1–20 (30.6%) and 22–30 (22%) were the digestion time most used. The most used dosage range of iron-based additives was between 0.11 and 1.0 (30.2%).

As presented in Fig. 1, the largest number of studies were

Table 1

Identification, levels and definition of moderators used in the analyses and percentage of case studies per moderator level.

Moderator	Levels	Definition	% distribution in the matrix
Published paper	Several	Identifies the primary studies	
		Complete references are given in supplementary section	
Iron-based Additive-type	Fe	Iron powder	5.3
	Fe NPs	Nanoparticles of iron powder	2.3
	Fe(NO) ₃	Iron nitrate	1.2
	Fe(OH) ₃	Iron (III) oxide-hydroxide	0.3
	Fe ₂ O ₃	Iron (III) oxide (Hematite)	2.6
	Fe ₂ O ₃ NPs	Iron (III) oxide nanoparticles	3.2
	Fe ₃ O ₄	Magnetite powder	4.1
	Fe ₃ O ₄ NPs	Magnetic iron oxide nanoparticles	18.8
	Fe ₄ NiO ₄ NPs	Ni ferrite Nanoparticles	1.2
	FeCl ₂	Iron (II) chloride	7.3
	FeCl ₃	Iron (III) chloride	6.7
	FeSO ₄	Iron (II)sulphate	4.1
	NVSI	Nano zerovalent scrap iron	1.2
	NZVI	Both commercial and synthesized NZVI	21.4
	Stainless steel	Alloy of iron and other elements	0.9
	WIS	Clean and rusty iron scrap	4.1
	ZVI	Zero valent iron powder	15.2
Substrate type (organic waste)	Cattle waste	Wastes from cattle dung, dairy manure and allied wastes	7.0
	combined organic waste	more than one organic wastes	8.2
	Food waste	Food and vegetables	8.5
	green algae	Entire algae wastes	0.6
	Pig manure	Wastes from piggery (urine and dung)	5.6
	Poultry litter	Solid, liquid and absorbents wastes	2.9
	Rice straw	Rice straws and husks	5.6
	Sludge	Waste activated sludge, anaerobic sludge etc.	58.1
	wastewater	Blackwater, brewery wastewater etc.	3.5
Effective digester volume (Batch)	80–300	Small laboratory bioreactors	33.7
	310–900	Moderate lab. reactors	43.1
	1000–10000	Large lab. and pilot size bioreactors	12.9
Semi-continuous	300–600	Moderate size reactors	0.6
	3000	Large lab. reactors	2.3
Continuous	240–400	Moderate lab. continuous reactors	3.5
	2000–12000	Large lab and pilot scale bioreactors	3.8
Dosages	0.00005–0.001		5.6
	0.01–0.00122		7.9
	0.015–0.1		21.1
	0.11–1.0		30.2
	1.1–10		27.0
	12–81.04		7.9
	300		0.3
Hydraulic Retention Time	1.5–20		30.6
	22–30		22.6
	31–60		35.5
	65–100		4.6
	20(80)-20(376)		6.7
Temperature	30–39	Mesophilic conditions	96.4
	45–55	Thermophilic conditions	3.6
Carbon–Nitrogen ratio	5.8–18.6	Low range	55.8
	20–30	Optimum range	18.6
	40–72.3	High range	25.7

conducted in China, 202 (59.2%), this may be due to favorable the policies, laws and subsidies on biogas technology research and development by the Chinese both at small and large scales (Zheng et al., 2020). Others study locations are Pakistan 32 (9.4%) studies, Japan, 26 (7.6%), USA and Iran, 17.(5%), Egypt, 9 (2.6), France, 6(1.8), Spain and Turkey, 5 (1.5%), Singapore, 4 (1.2%), Benin, Canada and South Korea, 3 (0.9%), Australia, Italy and Sweden, 2 (0.6%), India, Mexico and Netherland, 1 (0.3%). The study location with lower research contribution may have more research on iron additive enhancement but may have been published in other languages (not English which is part of the inclusion criteria in this study).

Studies in the database were published between 2013 and August 2019, as shown in Fig. 2, the largest publication was recorded in 2019 (32%) and 2018 (26%), the lowest was recorded in 2013 (1%). The continuous yearly rise in iron-based supplementation research may be attributable to the global quest to optimize

existing substrate sources or enhance new substrates with additives (Romero-Güiza et al., 2016). In the recent years, increasing research on additives supplementation for enhanced anaerobic digestion may also be due to favorable renewable energy policies (Zheng et al., 2020).

3.2. The overall effect of iron-based additive-s on biogas enhancement

Data from the 100 research articles included for the meta-analysis of iron-based additives impact on biogas yields from enhanced anaerobic digestion processes were tabulated and attached as Supplementary Table S1. The pooled estimate determined with REM revealed that the iron-based additive types, dosages, substrate type and digester types/volumes, as well as location and year of study significantly enhanced biogas yields

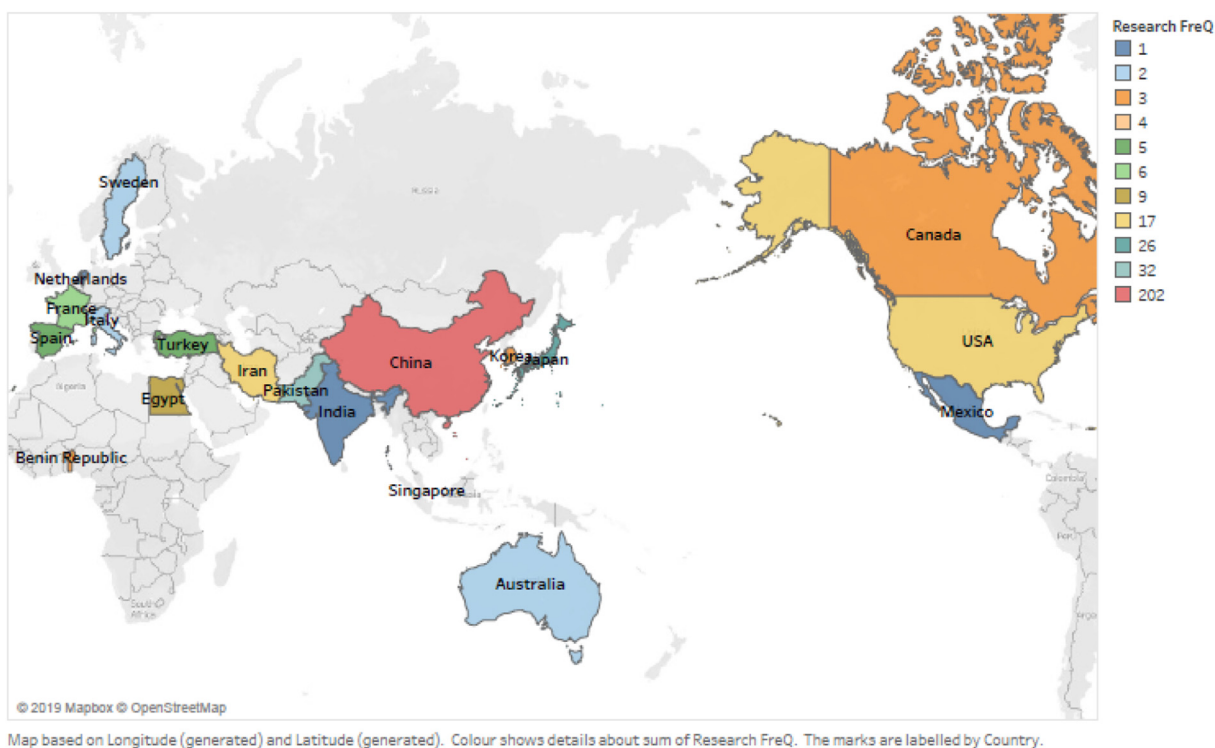


Fig. 1. Experiment locations included in meta-analysis. Map is based on generated longitude and latitude. Shaded color shows sum of research frequency worldwide. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

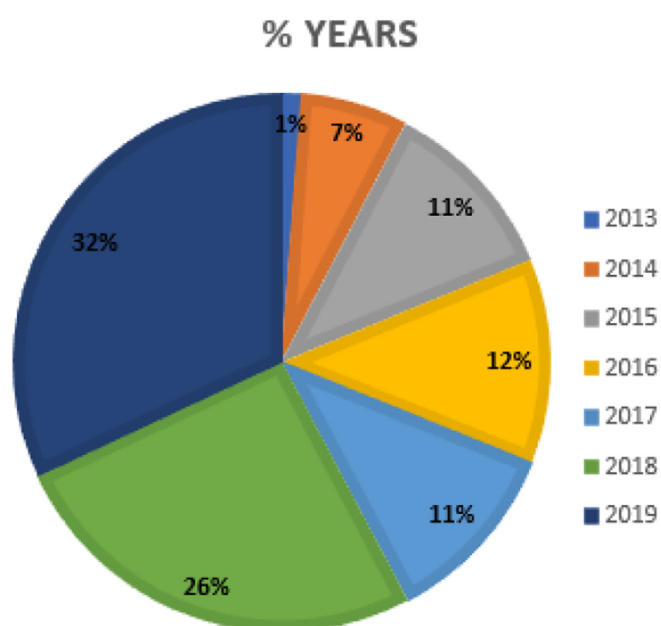


Fig. 2. Pie-charts for years of study articles included in the database.

(SMD = 1.451, 95% CI: 1.280–1.622, $p < 0.001$; [Supplementary Fig. S1](#) and [Figs. 3–5, 7, 10 and 11](#)). Temperature range (SMD = 1.487, 95% CI: 1.317 to 1.658, $p < 0.001$), HRT (SMD = 1.520, 95% CI: 1.346 to 1.694, $p < 0.001$) and C:N ratio (SMD = 1.663, 95% CI: 1.352 to 1.974, $p < 0.001$) were all significant and presented in [Figs. 6, 8 and 9](#). These significant enhancement results support the beneficial influence of iron-based supplementation in enhancing biogas yield. However, since biogas yield is an end-product means

of evaluating the performance of anaerobic digestion processes, the observed significant enhancements suggest improvement of methanogenic activities in comparison to that of the controls ([Ugwu and Enweremadu, 2019b](#); [Casals et al., 2014](#); [Angelidaki and Sanders, 2004](#)).

Similarly, the notable enhancement of the biogas yield by iron additive supplementation may be based on its ability to lose or gain electron during anaerobic digestion, aid enhancement of methanogens, support DNA replication, enhanced direct interspecies electron transfer (DIET), facilitate syntrophic methanogenic metabolism, etc. ([Zhang et al., 2019](#); [Cruz Viggi et al., 2014](#)). The overall enhancement agrees with previous reports that iron-based additives of different forms, types and optimum dosages improve biogas yield, achieve better biodegradability of organic substrates, get higher diminution of H_2S and more stabilize sludges ([Suanon et al., 2017](#); [Zhang et al., 2018](#)).

On the other hand, the heterogeneity test which is vital for evaluating sources of variation between studies in meta-analysis revealed significant evidence of heterogeneity ($Q = 848.281$, $p < 0.001$, $I^2 = 59.92$, [Fig. S1](#)). In determining the actual sources of the moderate heterogeneity observed in this meta-analysis, the variables (additive types, dosage range, substrate types, digester volume/type, study location, year of study, temperature, HRT and C: N ratio) were studied via sub-analysis and meta-regression.

3.3. Sub-analysis of covariates and their influences on biogas yield enhancement

The influences of factors identified in [Lohani and Havukainen \(2018\)](#) and [Hassanein et al. \(2019\)](#) such as additive types, substrate types, dosage ranges, C: N ratio, HRT, temperature ranges, effective digester volume/type, research location and year on iron-enhanced anaerobic digestion of organic wastes and biogas yield were evaluated and discussed at subgroup levels. The sub-analyses

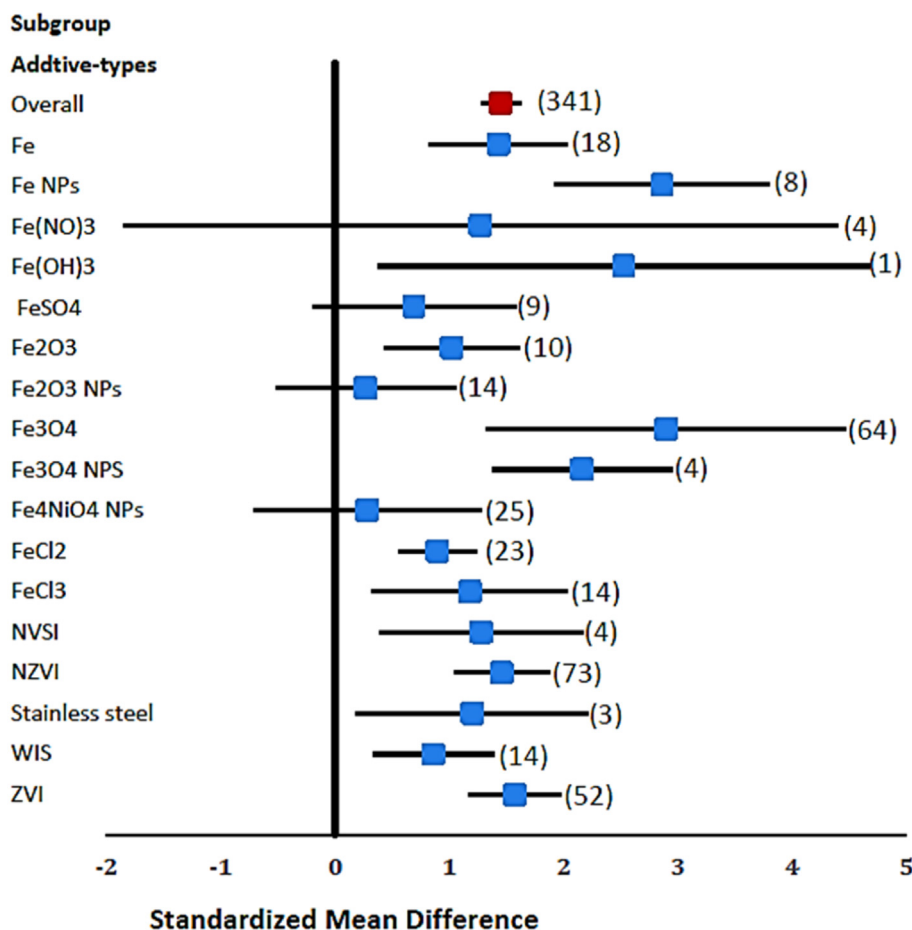


Fig. 3. Influence of different additive types on biogas yield effect sizes of enhanced anaerobic digestion with iron-based additives. The overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

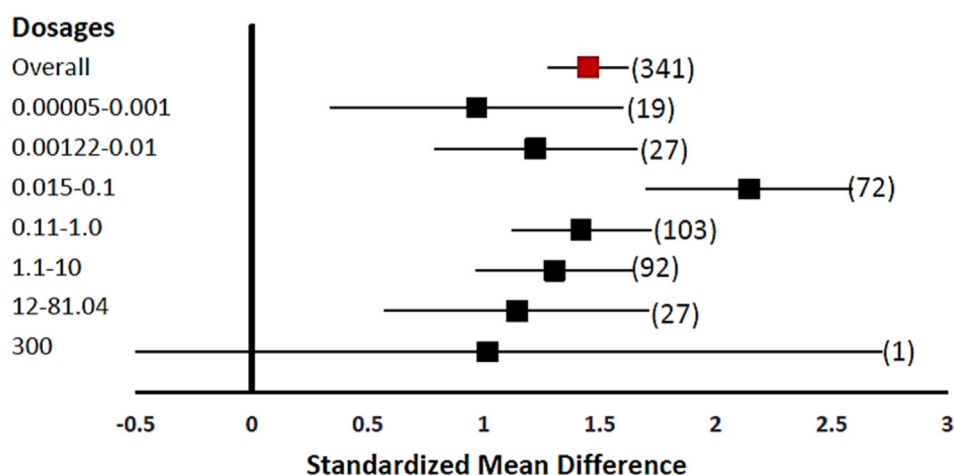


Fig. 4. Influence of different dosage ranges on biogas yield effect sizes of enhanced anaerobic digestion with iron-based additives. The overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

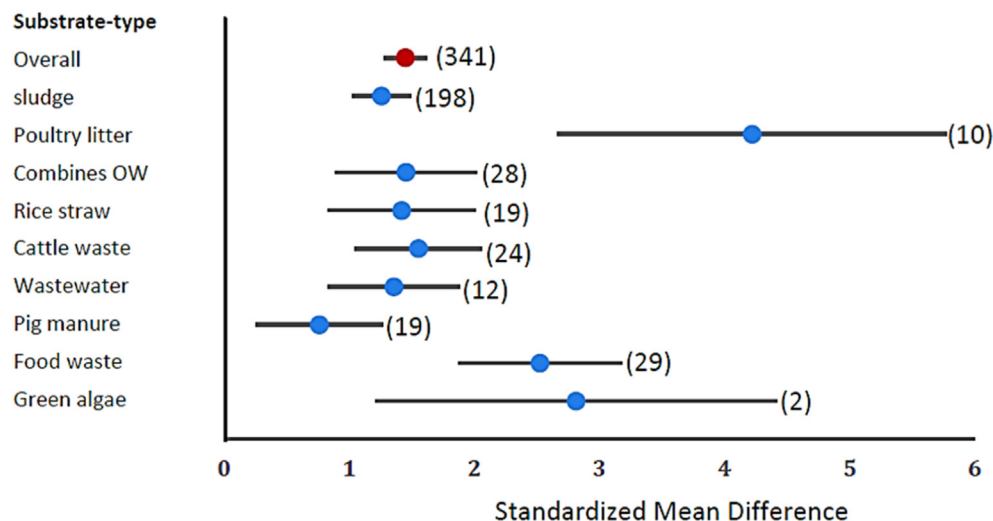


Fig. 5. Effects of substrate types on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

of the variables were analyzed to identify moderators that significantly affect biogas yield during anaerobic digestion of organic wastes.

3.3.1. Additive types

In this meta-analysis, most of the additive types included were largely significant in the enhancement of biogas yield. Additive type Fe_3O_4 (SMD = 2.899, 95% CI: 1.327 to 4.471, $p < 0.001$, Fig. 1) had the most significant influence on the biogas yield during anaerobic digestion processes, followed by Fe_3O_4 NPs, Fe and Fe NPs. The performances of these conductive iron additives (Fe_3O_4 and Fe_3O_4 NPs) and Fe NPs agreed with Zhao et al. (2017) that anaerobic digestion process supplemented with conductive iron yields requisite potential for enhancing decomposition of organics into simpler forms. This is done through the mechanisms of dissimilatory iron reduction and DIET promotion for syntrophic metabolism of volatile fatty acids (VFAs). In Casals et al. (2014), the highest biogas yield (180%) per gram of organic matter ever reported was attained with the addition of Fe_3O_4 NPs to an anaerobic digestion process. In agreement with the better performance of conductive iron-based additives, Abdelsalam et al. (2017), reported higher biogas yield from Fe_3O_4 NPs than yields from NZVI. Similarly, Hassanein et al. (2019) reported that Fe NPs enhanced biogas yield more than other nanoparticles used.

However, Fe_2O_3 NPs additive (SMD = 0.267, 95% CI: 0.518 to 1.053, $p = 0.505$, Fig. 3) with the least SMD, was not significant in biogas enhancement, which concurs with Ambuchi et al. (2016) that the use of Fe_2O_3 NPs resulted in slower degradation stimulation and lower biogas production. $\text{Fe}(\text{NO})_3$, FeSO_4 and Fe_4NiO_4 NPs were some of the additive types that had no significant effect on biogas enhancement. $\text{Fe}(\text{OH})_3$ had a low sample size, hence its level of the association was not determined. Generally, additive types had overall significant influence on biogas yield during anaerobic digestion of organic wastes enhanced with iron-based additives.

3.3.2. Dosage ranges

The attainment of optimum dosage range is the focus of every enhancement process. According to Casals et al. (2014), maintenance of optimum iron dosage during anaerobic digestion is very vital for sustained activities of anaerobes. In this study, all the

additive dosage ranges (0.00005–0.001, 0.00122–0.01, 0.015–0.1, 0.11–1.0, 1.1–10, 12–81.04, ≥ 300 g) significantly enhanced biogas yields. Dosage range of 0.015–0.1 g (SMD = 2.144, 95% CI: 1.699 to 2.589, $p < 0.001$, Fig. 4) had the most significant dosage advantage over other ranges included in the meta-analysis. This range is in line with the findings of Abdelsalam et al. (2017), where it was reported that 20 mg (0.02 g/L) of both NZVI and Fe_3O_4 NPs outclassed other dosages reported.

Similarly, Ali et al. (2017) recommended that an optimum dose (75 mg/L) of Fe_3O_4 achieved the maximum biogas yield. In Hassanein et al. (2019), it was observed that all lower doses (<0.1 g/L) of nanoparticles gave higher biogas yield, but 15 mg/L of Fe NPs achieved the highest yield. Although iron-based additives aid the anaerobic digestion process, excess dose of it inhibits methanogenic activities due to its contention for electrons with methanogens during electron transfer processes. High concentrations of ions of iron cause toxicity and kill anaerobes (Zhao et al., 2018; Casals et al., 2014). According to Puyol et al. (2018), excess dosage encourages complex formation like vivianite, siderite etc., increasing operational costs in industries and polluting the environment. The level of association of the dosage ≥ 300 g was not available because of low sample size.

3.3.3. Substrate type and C: N ratio

Substrate types, which are organic feedstocks (biomass) has cellulose, hemicellulose, carbohydrates, proteins and fats as its main constituents (Akunna, 2018). Biodegradability of substrates are influenced by iron-based additives in anaerobic condition (Amen et al., 2017; Abdelsalam et al., 2017). In this study, the restricted sub-analysis revealed that all substrate types significantly affected biogas yield of iron-additive enhanced digestion process (Fig. 5). Unlike in pig waste streams with the lowest observed yield, poultry litters showed the highest positive effect on biogas yield of iron enhanced anaerobic digestion (SMD = 4.221, 95% CI: 2.664 to 5.778, $p < 0.001$, Fig. 5), followed by algae biomass. This is in harmony with the reports of Hassanein et al. (2019) and Aguilar-Moreno et al. (2020) which showed that Fe NPs and Fe_3O_4 NPs achieved great yields upon enhancement of poultry wastes.

Likewise, the high effect of algae biomass on biogas yield agrees with the report of Zhu et al. (2017), that the presence of lipids in

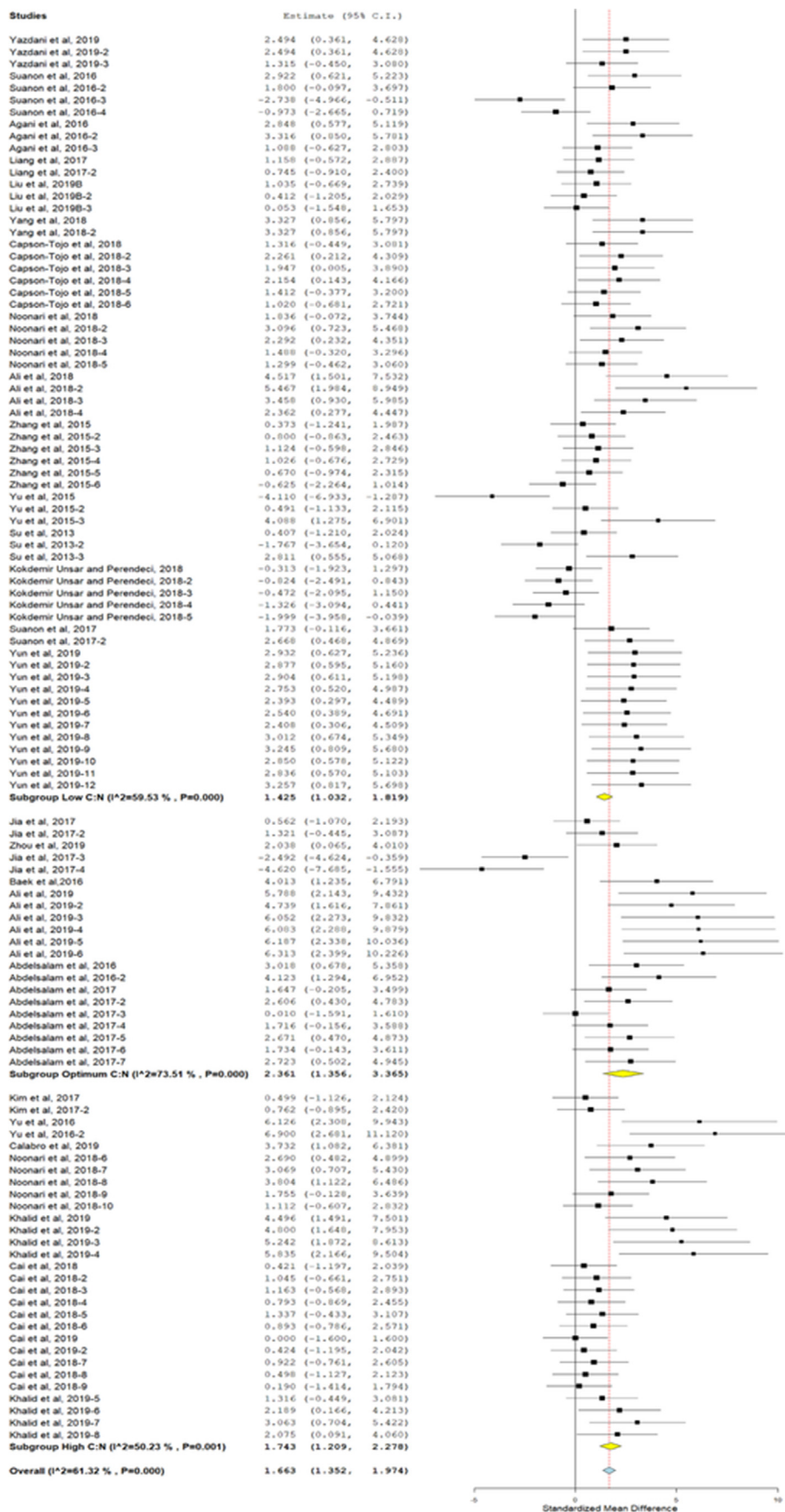


Fig. 6. Effect of C:N ratio on biogas yield of enhanced anaerobic digestion with iron-based additives. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size.

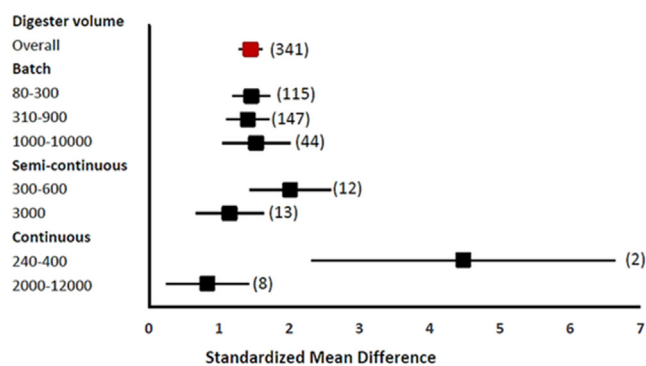


Fig. 7. Effects of digester type and effective volume on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

algae biomass gave rise to high biogas yield attainment. Just as it was observed in this study, [Li et al. \(2013\)](#) revealed that lignocellulosic substrate types had lower biogas yield and biodegradability. The variation in biogas contents and compositions produced during anaerobic digestion is because substrates differ from each other in origins, contents and compositions ([Raposo et al., 2011](#)).

On the other hand, the C: N ratio is an important factor in substrates that represents the carbon and nitrogen relationship. The sub-analysis of substrate's C: N ratios for enhanced anaerobic digestion with iron-based additives showed that all (low, optimum and high) C: N ratios were found highly significant. The optimum C:N ratio had the greatest enhancement (SMD = 2.361, 95% CI: 1.356 to 3.365, $p < 0.001$, [Fig. 6](#)), this agrees with [EPA \(2012\)](#); [Ugwu and Enweremadu \(2019\)](#) that substrates with C: N ratio between 20 and 30 produces largest amount of biogas when compared to those with lower and higher C: N ratios.

Since low C: N ratio indicates high protein content, resulting in increased free ammonia, which raises the pH and inhibits methanogenic activities ([Jingura and Kamusoko, 2017](#)), therefore it is difficult to achieve high biogas yield with low C: N substrates. Similarly, high C: N ratio of lignocellulosic substrate type is the main limiting factor for enhanced biogas yield during anaerobic digestion, it elicits fast diminution of nitrogen and causes low biogas production ([Eduok et al., 2018](#); [Jingura and Kamusoko, 2017](#)). [EPA \(2012\)](#) and [Eduok et al. \(2018\)](#) suggested that the synergistic approach of combining substrates of low C: N and high C: N ratios (co-digestion) enhances biogas yield, thereby increasing profitability in biogas firms.

3.3.4. Effective bioreactor volume/types

Bioreactors for anaerobic digestion exist in various configurations and are broadly categorized into the batch, semi-continuous and continuous as well as based on its capacities (volumes). The entire biodigester types/effective volumes were statistically significant to the influence of iron-based additives supplementation on biogas yield during the subgroup analysis. Generally, the largest increase was seen in the continuous digester types 240–400 ml (SMD = 4.483, 95% CI: 2.320 to 6.646, $p < 0.001$, [Fig. 7](#)), followed by semi-continuous 300–600 (SMD = 2.016, 95% CI: 1.436 to 2.595, $p < 0.001$, [Fig. 7](#)) and batch 1000–10000 ml (SMD = 1.533, 95% CI: 1.053 to 2.013, $p < 0.001$, [Fig. 7](#)). This is in sync with [Miranda et al. \(2016\)](#), which reported higher yield from continuous biodigester

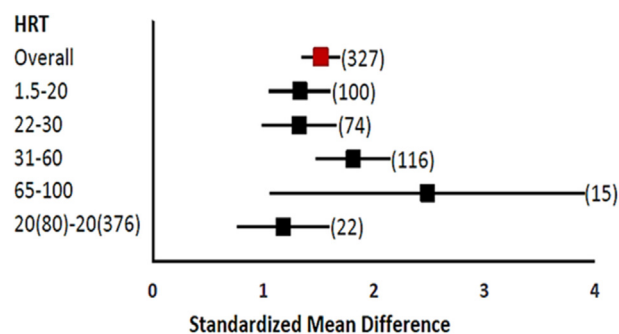


Fig. 8. Effects of HRT on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

type and semi-continuous type and suggested that the low influence seen in the batch reactors may be because the volumes are small and iron additives added get used up.

It was observed in this study that higher volumes of the batch reactor type (greater than 1000 ml) had a higher influence on biogas yield, this was consistent with the findings in [Miranda et al. \(2016\)](#). According to [Raposo et al. \(2011\)](#), the nature of substrates determines the effective volume and more homogeneous substrates necessitate the use of smaller reactor volumes. [bib_Khoufi_et_al_2015](#) Also, [Khoufi et al. \(2015\)](#) reported that biogas yield from semi-continuous bioreactor type was better enhanced than that from the batch anaerobic reactor.

3.3.5. Hydraulic retention time and temperature ranges

Hydraulic retention time (HRT) is an essential bioreactor operational factor, which influences volatile solids conversion into biogas during anaerobic digestion process as well as influences biodegradation of cellulose, hemicellulose, and crystalline cellulose ([Shi et al., 2017](#)). In the sub-analysis, anaerobic digestion processes enhanced with iron-based additives had overall significant influences on the HRTs. The HRT range of between 65 and 100 days (SMD = 2.486, 95% CI: 1.065 to 3.907, $p < 0.001$, [Fig. 8](#)) had the greatest observed impact, closely followed by HRT range of 31–60 days. While agreeing with this significant influence, [Romero-Güiza et al. \(2016\)](#), stated that additive supplementation influenced the HRT and achieved higher biogas production. [Shi et al. \(2017\)](#) indicated that although most lignocellulosic substrates need long HRT for substantial degradation, shorter HRT is most desirable for cost reduction and process efficiency increase.

More so, the temperature of bioreactors contributes immensely in influencing biodegradation and microbial activities of iron-supplemented anaerobic digestion process ([Romero-Güiza et al., 2016](#)). In analyzing effect size in relation to the influence of digestion temperature, overall positive and significant influence on the biogas yield of an iron-based supplemented anaerobic digestion was observed. The thermophilic temperature range showed no significant effect, but significant positive impacts were observed at mesophilic (SMD = 1.495, 95% CI: 1.326 to 1.665, $p < 0.001$, [Fig. 9](#)). This trend agrees with [Zhang et al. \(2018b\)](#) that addition of ZVI at mesophilic temperature improved biogas yield by 4.7%, while the yield increase at thermophilic temperature yield increase was 3.9%.

Similarly, [Wang et al. \(2019b\)](#) revealed that the 5 g/L NZVI supplementation at mesophilic temperature range achieved the

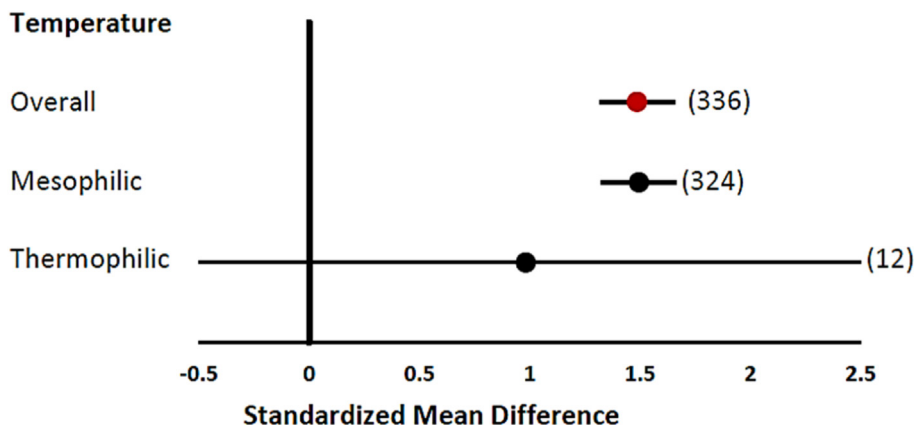


Fig. 9. Effects of temperature ranges on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

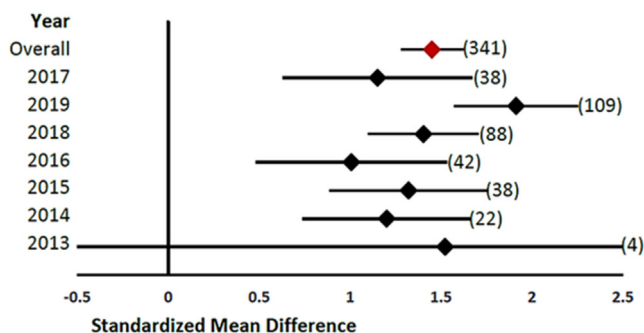


Fig. 10. Effects of Year of study on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

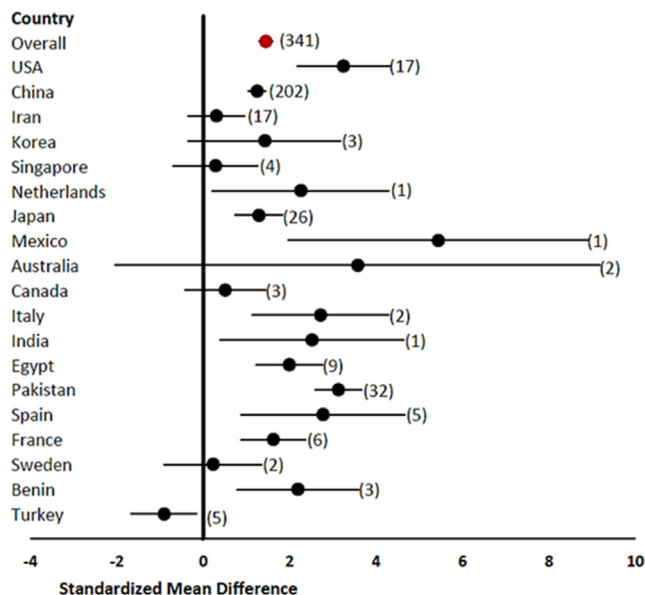


Fig. 11. Effects of Country of study on biogas yield, the overall effect size for the sub-analysis performed in this study are highlighted in red. The error bar indicates the 95% confidence interval (CI) and the vertical continues line (effect size = 0) indicate no effect. The effect Mean effect size >0 indicates increase, while mean effect size <0 indicates inhibition decrease due to iron-based supplementation. The effect is significant when 95% CI does not overlap 0, but moderator levels do not differ significantly when 95% CI overlap. Values in the parentheses depict sample size. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

maximum biogas yield than the yield at thermophilic temperature. According to Romero-Güiza et al. (2016), the thermophilic anaerobic digestion requires higher additives owing to the decrease in additive bioavailability or increased additive demands. Therefore, the low influence of iron-enhanced biogas yields under thermophilic conditions in this study was because about 89.7% of the bioreactors reported were batch types and further additive demands after the initial use-up are not met. Generally, the use of iron-based additives enables bioreactor operation at higher organic loading rate, lower temperature demand, lower HRT and ultimately higher biogas yield Schmidt et al. (2014) and Romero-Güiza et al. (2016). This implies that the use of iron-based supplements at mesophilic temperature ranges aids reduction in industrial operational costs for further heating.

3.3.6. Study year and location

The result of the subgroup analysis for the study years was highly significant to the influence of iron-based supplementation on the biogas yield. Aside studies conducted in 2013, which was not significant (SMD = 1.523, 95% CI: 1.160 to 4.206, $p = 0.266$, Fig. 10), all others study years were highly significant. 2019 had the most significant influence on the biogas yield enhanced with iron-based

additives (SMD = 1.913, 95% CI: 1.571 to 2.254, $p < 0.001$, Fig. 10), indicating that iron supplementation of anaerobic digestion process is on the rise yearly. This yearly rise may be attributable to yield maximization drive of firms, dwindling substrate source and favorable policies (Zheng et al., 2020; Ugwu and Enweremadu, 2019b).

More so, from the performance of the restricted sub-analysis, significant enhancement was observed between the biogas yields and the research carried out in the USA, China, Japan, Italy, Egypt, Pakistan, Spain, France and Benin, while no association existed between biogas enhancement and studies located in Iran, South Korea, Singapore, India and Sweden showed no significance.

Inhibition was noticed in studies from Turkey (SMD = -0.904 , 95% CI: 1.669 to -0.138 , $p = 0.021$, Fig. 11). Although there has not been any study on the impact of location variations on iron-based enhanced biogas production, but it has been reported that location and climatic variations, agronomic practices influence biogas yield (Raposo et al., 2011).

Jingura and Kamusoko (2017) also reported that factors associated with locations (agro-climatic conditions, soil characteristics and agronomic practices) affect the waste types and substrate compositions, then influence biogas yields.

However, large number of reported researches on iron supplementation of anaerobic digestion in countries like china may be due to favorable governmental policies, laws and subsidies on renewable energy research and development (Zheng et al., 2020). Researchers, industries and policy makers may be able to access advanced iron enhancement technologies and information than in other locations.

3.4. Effects of moderators and interactions on the effect sizes of biogas yield

Meta-regression of the influence of iron-based additives on biogas yield during anaerobic digestion of organic wastes ($Q_B = 29$, $df = 1$, $p < 0.001$ and $Q_B = 340$, 848.28, $p < 0.001$, Fig. S1) revealed that additive types ($Q_B = 36.192$, $df = 16$, $p = 0.003$), substrate type ($Q_B = 35.534$, $df = 8$, $p < 0.001$), location of study ($Q_B = 96.053$, $df = 18$, $p < 0.001$) and dosage range ($Q_B = 12.027$, $df = 6$, $p = 0.05$) ranges were the significant variables describing the variation and referred to as the predictors of effects.

The mixed-effects meta-regression analysis revealed the differences among cumulative effect sizes of the various groups. It showed that year of study, C:N ratio, temperature, biodigester type/effective volume and HRT were not significant explanatory variables ($p = 0.07$, $p = 0.193$, $p = 0.182$, $p = 0.158$ and $p = 0.233$ respectively). Analysis of variance suggests that 59.88% of the total variance was between studies ($I^2 = 59.8\%$), and the covariates could explain 47.72% of the between-study variance ($R^2 = 47.72\%$). In line with the subgroup analysis results, meta-regression confirmed that significant covariates were sensitive to enhanced biogas yield upon the addition of iron-based additives. However, the determination of active relationship between additive types and biogas yield showed that the regression coefficient at $p < 0.05$ were 2.098 and 0.846 respectively for Fe_3O_4 and Fe_2O_3 . Ranges of mesophilic and thermophilic temperatures were not statistically different at $p < 0.05$, indicating that temperature may not be a very significant predictor.

Furthermore, the influence of interaction between the significant covariates and the treatment was observed and shown in Table 2. From the result, significant interactions at 95% CI were noticed across most of the predictors. Significant interaction indicates the dependence of each covariates on the magnitude of each other (Borenstein et al., 2009; Wallace et al., 2017). The highest interaction influence of 5.414 and 4.009 were respectively observed with the additive types-dosage and dosage-HRT interaction respectively, indicating that each additive type has an optimum dosage for maximum yield and optimum dosage affects the HRT (Romero-Güiza et al., 2016). Additive-dosage interaction was able to explain 25.19% of its heterogeneity while dosage-HRT explained 33.97% of the noticeable heterogeneity. More so, the strong interaction effects between additive types and all other covariates, justified the views in Romero-Güiza et al. (2016) that various additive types affect biogas yield differently.

The meta-regression showed that the study year was not a predictor, its treatment-predictors interaction with location had the least interaction effect, it also showed weak interactions effects with other predictors. However, a significant treatment-predictor

interaction effect between substrate types and all moderators was observed, which agreed with Angelidaki and Sanders (2004) that substrate depends on other variables at optimum levels for biogas production. While agreement with its significant interaction with study location (which is also a predictor), Mayer et al. (2014) stated that substrate location was accountable for most of the dissimilarity of the biogas yield from other locations.

3.5. Cumulative meta-analysis

Cumulative meta-analysis was conducted with the included dataset of biogas yields enhanced by iron-based additives. This revealed the temporal changes in the magnitude and stability of effects reported (Fig. S2). In comparing the control and additive-enhanced biogas yields, all studies (2013–2019) showed positive effect sizes, indicating superior and significant additive enhancement of biogas yield. There tend to be relatively stabilized effect sizes as the number of studies and the diversity of additive types increased from 2015, more unstable and non-significant effect sizes were observed in studies between 2013 and 2014. The magnitude of the mean effect sizes reduced between 2013 and 2015, thereafter, it became more significant and increased. It was also observed that the mean effect sizes got stabilized from 2016 to 2019. Also, the variance around the mean effect notably fell, revealing enhancement of biogas yield upon addition of iron-based additives. In conclusion, some studies conducted in 2013 had insignificant mean effect size but turned significant with the addition of more studies from other years, hence ensuring an overall stable and significant effect size (Fig. S2).

3.6. Sensitivity analysis

A sensitivity analysis was performed to determine the effectiveness of each study on the overall estimated effect size. In this study, four studies (Ali et al., 2019; Jia et al., 2017; Lizama et al., 2019; Yu et al., 2016) with effect sizes of 15 were excluded. The pooled effect upon exclusion of individual studies did not significantly alter the magnitude and direction when compared to the use of the whole sample sizes (Fig. S1) but reduced the heterogeneity value ($Q = 753.92$, $p < 0.001$, $I^2 = 56.89$). The influence of iron-based additive supplementation on biogas yield did not significantly change with the removal of the four studies, indicating that supplementation of iron-based additives to anaerobic digestion of organic wastes, improves biogas yield (SMD = 1.384, 95% CI: 1.218 to 1.550, $p < 0.001$). This goes to show that the result of this meta-analysis can be interpreted with a high level of certainty.

3.7. Analysis of publication bias

For the analysis of the publication bias, the Rosenberg fail-safe number (Nfs) for the entire database is 42452, which is 24.8-fold above the threshold of 1715 ($5 \times n + 10$) needed to consider the mean effect size robust. For moderators, the Nfs of additive-type, substrate-type, country, year, dosage range and biodigester volume/type are the same with that of the entire database, but that of temperature (43903), HRT (42863) and C: N ratio (11345) are 25.9, 26.05 and 19.7 folds above the threshold of 1690 ($5 \times n + 10$), 1645 ($5 \times n + 10$) and 575 ($5 \times n + 10$) respectively. However, publication bias was not a source of concern in this study, a relatively big amount of unpublished studies would be needed to alter the significant effects statistically. There was no evidence of publication bias from analysis of the influence of iron-based additives on biogas yield.

Table 2
Interaction influences of treatment-predictors with covariates.

Predictors	Interacting covariate	wES	Q	R ² (%)	P value
Additive types	C: N	2.046	24.63	53.08	1.79E-03*
	Country	−2.904	40.3	43.78	2.32E-04*
	Dosages	5.414	69.5	25.19	0.000136 *
	HRT	3.176	77.77	21.76	4.61E-07*
	Substrate type	1.232	96.9	53.65	8.29E-10*
	Temperature	1.242	12.141	5.29	0.0163*
	Digester volume	1.240	65.4	20.76	3.01E-05 *
	Year	−380.112	27.48	13.82	0.00659*
Dosage	C: N	1.413	13.4	12.08	0.0638 ^{NS}
	Country	−0.313	47.6	45.14	4.80E-04*
	HRT	4.009	95.2	33.97	2.73E-13*
	Substrate type	1.008	40.85	18.32	9.81E-04*
	Temperature	1.382	7.17	0.68	0.0668 ^{NS}
	Digester volume	1.829	22.73	4.44	0.15822 ^{NS}
	Year	−1140	16.43	5.49	0.00572*
	C: N	0.588	32.19	52.9	5.45E-06*
Substrate Type	Country	−0.945	18.7	46.43	4.69E-03*
	HRT	1.029	74.88	39.08	3.87E-11*
	Temperature	1.267	13.55	14.12	1.14E-03*
	Digester volume	1.108	49.0	31.02	4.05E-07 *
	Year	−52.916	22.4	19.19	0.001008*
	C: N	2.041	19.72	61.89	1.94E-04*
	HRT	1.77	27.61	36.79	1.11E-04*
	Temperature	1.27	81.74	26.89	1.88E-10*
Study location	Digester volume	1.032	21.7	40.08	2.87E-03*
	Year	−140.803	24.82	39.92	3.17E-03*

wES = Weighted mean Effect Size of the interactions. P value = probability level at 95%.
R² = Amount of heterogeneity accounted for (%). Q = Test of Moderators.

4. Conclusion

The results of the entire pooled estimates of the included research publications in the meta-analysis revealed that addition of iron-based additives is significantly associated with enhancement of biogas yield in the anaerobic digestion of organic biomasses. It was established in this research that optimum C:N ratio was the best range suited for biogas yield in an enhanced anaerobic digestion process. It can be inferred that when anaerobic digestion process is enhanced with iron-additives, the mesophilic temperature range yields higher biogas than when performed under thermophilic conditions. Based on the results from this study, anaerobic digestion process of organic waste with conductive iron (Fe₃O₄ and Fe₃O₄ NPs) additive types of dosage between 0.015 and 0.1 g/L, under mesophilic condition in a continuous bioreactor of effective volume between 240 and 400 mL is expected to achieve optimum yield. The largely significant treatment-predictor interactions between-covariates (especially with additive types), showed that iron-supplementation of anaerobic digestion processes enhances biogas yield.

In a nutshell, the evidence-based information on iron enhanced anaerobic digestion with optimum inputs and operational parameters (moderators) from this study will aid the achievement of environmental sustainability through higher substrate degradability and waste reduction. It is also expected to ultimately increase biogas yield at reduced production cost and guide policy makers in the biogas industries. Further exploration of additive supplementation of anaerobic digestion processes for enhanced biogas yield with multi-factor and other meta-analytic options is recommended.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

S.N. Ugwu: Conceptualization, Methodology, Formal analysis, Writing - original draft. **R.K. Bischoff:** Writing - review & editing, Visualization. **C.C. Enweremadu:** Supervision, Conceptualization, Writing - review & editing.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2020.122449>.

References

- Abdelsalam, E., Samer, M., Attia, Y.A., Abdel-Hadi, M.A., Hassan, H.E., Badr, Y., 2017. Influence of zero valent iron nanoparticle and magnetic iron oxide nanoparticle on biogas and methane production from anaerobic digestion of manure. *Energy* 120, 842–853. <https://doi.org/10.1016/j.energy.2016.11.137>.
- Adu, M.O., Yawson, D.O., Armah, F.A., Asare, P.A., Frimpong, K.A., 2018. Meta-analysis of crop yields of full, deficit and partial root-zone drying irrigation. *Agric. Water Manag.* 197, 79–90. <https://doi.org/10.1016/j.agwat.2017.11.019>.
- Adu, M.O., Yawson, D.O., Abano, E.E., Asare, P.A., Armah, F.A., Opoku, E.K., 2019. Does water-saving irrigation improve the quality of fruits and vegetables? Evidence from meta-analysis. *Irrigation sci* 37, 669–690. <https://doi.org/10.1007/s00271-019-00646-2>.
- Aguilar-Moreno, G.S., Navarro-Ceron, E., Velazquez-Hernandez, A., Hernandez-Eugenio, G., Aguilar-Mendez, M.A., Espinosa-Solares, T., 2020. Enhancing methane yield of chicken litter in anaerobic digestion using magnetite nanoparticles. *Renew. Energy* 147, 204–213. <https://doi.org/10.1016/j.renene.2019.08.111>.
- Akunna, J.C., 2018. *Anaerobic Waste-Wastewater Treatment and Biogas Plants: a Practical Handbook*. CRC Press, Taylor & Francis Group, Boca Raton, FL.
- Ali, A., Mahar, R.B., Soomro, R.A., Sherazi, S.T.H., 2017. Fe₃O₄ nanoparticles facilitated anaerobic digestion of organic fraction of municipal solid waste for

- enhancement of methane production. *Energy Sources Part A* 39 (16), 1815–1822. <https://doi.org/10.1080/15567036.2017.1384866>.
- Ali, A., Mahar, R.B., Abdelsalam, E.M., Sherazi, S.T.H., 2019. Kinetic modeling for bioaugmented anaerobic digestion of the organic fraction of municipal solid waste by using Fe₃O₄ nanoparticles. *Waste Biomass Valor* 10 (11), 3213–3224. <https://doi.org/10.1007/s12649-018-0375-x>.
- Amen, T.W.M., Eljamal, O., Khalil, A.M.E., Sugihara, Y., Matsunaga, N., 2018. Methane yield enhancement by the addition of new novel of iron and copper-iron bimetallic nanoparticles. *Chem. Eng. Process* 130, 253–261. <https://doi.org/10.1016/j.cep.2018.06.020>.
- Amen, T.W.M., Eljamal, O., Khalil, A.M.E., Matsunaga, N., 2017. Biochemical methane potential enhancement of domestic sludge digestion by adding pristine iron nanoparticles and iron nanoparticles coated zeolite compositions. *J. Environ. Chem. Eng.* 5, 5002–5013. <https://doi.org/10.1016/j.jece.2017.09.030>.
- Ambuchi, J.J., Zhang, Z., Feng, Y., 2016. Biogas enhancement using iron oxide nanoparticles and multi-wall carbon nanotubes. *Int. schola. Sci. Res. Inno.* 10 (10), 2016. [Scholar.waset.org/1307-6892/10005619](https://scholar.waset.org/1307-6892/10005619).
- Angelidaki, I., Sanders, W., 2004. Assessment of the anaerobic biodegradability of macropollutants. *Rev. Environ. Sci. Biotechnol.* 3, 117–129. <https://doi.org/10.1007/s11157-004-2502-3>.
- Angelidaki, I., Alves, M., Bolzonella, D., Borzacconi, L., Campos, J.L., Guwy, A.J., Kalyuzhnyi, S., Jenicek, P., van Lier, J.B., 2009. Defining the biomethane potential: proposed protocol for batch assays. *Water Sci. Technol.* 59 (5), 927–934. <https://doi.org/10.2166/wst.2009.040>.
- Baredar, P., Suresh, S., Kumar, A., Krishnakumar, P., 2016. A review on enhancement of biogas yield by pre-treatment and addition of additives. *MATEC Web of Conferences* 62 (6002), 1–5. <https://doi.org/10.1051/mateconf/20166206002>.
- Bharathiraja, B., Sudharana, T., Jayamuthunagai, J., Praveenkumar, R., Chozhavadhan, S., Iyyappan, J., 2018. Biogas production – a review of composition, fuel properties, feedstock and principles of anaerobic digestion. *Renew. Sustain. Energy Rev.* 9, 570–582. <https://doi.org/10.1016/j.rser.2018.08.010>.
- Borenstein, M., Hedges, L.V., Higgins, J.P.T., Rothstein, H.R., 2009. *Introduction to Meta-Analysis*. John Wiley & Sons, Ltd, 2009.
- Cao, Y., Wang, X., Bai, Z., Chadwick, D., Misselbrook, T., Sommer, S.G., Qin, W., Lin Ma, L., 2019. Mitigation of ammonia, nitrous oxide and methane emissions during solid waste composting with different additives: a meta-analysis. *J. Clean. Prod.* 235, 626–635. <https://doi.org/10.1016/j.jclepro.2019.06.288>.
- Casals, E., Barrena, R., García, A., González, E., Delgado, L., Busquets-Fitè, M., Font, X., Arbiol, J., Glatzel, P., Kvashnina, K., Sánchez, A., Puentes, V., 2014. Programmed iron oxide nanoparticles disintegration in anaerobic digesters boosts biogas production. *Small* 10, 2801–2808. <https://doi.org/10.1002/sml>.
- Cohen, J., 1988. *Statistical Power Analysis for the Behavioral Sciences*, second ed. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Cruz Viggi, C., Rossetti, S., Fazi, S., Paiano, P., Majone, M., Aulenta, F., 2014. Magnetite particles triggering a faster and more robust syntrophic pathway of methanogenic propionate degradation. *Environ. Sci. Technol.* 48, 7536–7543. <https://doi.org/10.1021/es5016789>.
- DerSimonian, R., Laird, N., 1986. Meta-analysis in clinical trials. *Contr. Clin. Trials* 7, 177–188. [https://doi.org/10.1016/0197-2456\(86\)90046-2](https://doi.org/10.1016/0197-2456(86)90046-2).
- Eduok, S., John, O., Ita, B., Inyang, E., Coulon, F., 2018. Enhanced biogas production from anaerobic Co-digestion of lignocellulosic biomass and poultry feces using source separated human urine as buffering agent. *Front Environ Sci* 6, 67. <https://doi.org/10.3389/fenvs.2018.00067>.
- EPA, 2012. Increasing Anaerobic Digester Performance with Co-digestion. www.epa.gov/sites/production/files/2014-12/documents/codigestion.pdf. (Accessed 12 September 2019), accessed.
- Gao, N., Zhang, L., Wu, C., 2018. Biomass and wastes for bioenergy: thermochemical conversion and biotechnologies. *BioMed Res. Int.* 9638380, 1–2. <https://doi.org/10.1155/2018/9638380>.
- Hao, X., Wei, J., van-Loosdrecht, M.C.M., Cao, D., 2017. Analysing the mechanisms of sludge digestion enhanced by iron. *Water Res.* 117, 58–67. <https://doi.org/10.1016/j.watres.2017.03.048>.
- Hassanein, A., Lansing, S., Tikekar, R., 2019. Impact of metal nanoparticles on biogas production from poultry litter. *Bioresour. Technol.* 275, 200–206. <https://doi.org/10.1016/j.biortech.2018.12.048>.
- Hedges, L.V., Olkin, I., 1985. *Statistical Methods for Meta-Analysis*. Academic Press, Orlando, Florida.
- Higgins, J.P.T., Deeks, J.J., 2011. Chapter 7: selecting studies and collecting data (updated March 2011). In: Higgins, J.P., Green, S. (Eds.), *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0. The Cochrane Collaboration, 2011. Available from: <http://handbook.cochrane.org/>. (Accessed 10 August 2019) (Accessed on).
- Higgins, J.P.T., Thompson, S.G., 2002. Quantifying heterogeneity in a meta-analysis. *Stat. Med.* 21, 1539–1558. <https://doi.org/10.1002/sim.1186>.
- Hiloidhari, M., Das, D., Baruah, D.C., 2014. Bioenergy potential from crop residue biomass in India. *Renew. Sustain. Energy Rev.* 32, 504–512. <https://doi.org/10.1016/j.rser.2014.01.025>.
- Holliger, C., Alves, M., Andrade, D., Angelidaki, I., Astals, S., Baier, U., et al., 2016. Towards a standardization of biomethane potential tests. *Water Sci. Technol.* 74 (11), 2515–2522. <https://doi.org/10.2166/wst.2016.336>.
- Jia, T., Wang, Z., Shan, H., Liu, Y., Gong, L., 2017. Effect of nanoscale zero-valent iron on sludge anaerobic digestion. *Resour. Conserv. Recycl.* 127, 190–195. <https://doi.org/10.1080/09593330.2019.1604811>.
- Jingura, R.M., Kamusoko, R., 2017. Methods for determination of biomethane potential of feedstocks: a review. *Biofuel Res. J.* 14, 573–586. <https://doi.org/10.18331/BRJ2017.4.2.3>.
- Khoufi, S., Louhichi, A., Sayadi, S., 2015. Optimization of anaerobic co-digestion of olive mill wastewater and liquid poultry manure in batch condition and semi-continuous jet-loop reactor. *Bioresour. Technol.* 182, 67–74. <https://doi.org/10.1016/j.biortech.2015.01.092>.
- Kim, M., Li, D., Choi, O., Sang, B., Chiang, P.C., Kim, H., 2017. Effects of supplement additives on anaerobic biogas production. *Kor. J. Chem. Eng.* 34 (7), 1–8. <https://doi.org/10.1007/s11814-017-0175-1>.
- Lavergne, C., Jeison, D., Ortega, V., Chamy, R., Donoso-Bravo, A., 2018. A need for a standardization in anaerobic digestion experiments? Let's get some insight from meta-analysis and multivariate analysis. *J. Environ. Manag.* 222, 141–147. <https://doi.org/10.1016/j.jenvman.2018.05.030>.
- Lean, I.J., Rabiee, A.R., Duffield, T.F., Dohoo, I.R., 2009. Invited review: use of meta-analysis in animal health and reproduction: methods and applications. *J. Dairy Sci.* 92, 3545–3565. <https://doi.org/10.3168/jds.2009-2140>.
- Li, W., Khalid, H., Zhu, Z., Zhang, R., Liu, G., Chen, C., 2018. Methane production through anaerobic digestion: participation and digestion characteristics of cellulose, hemicellulose and lignin. *Appl. Energy* 226, 1219–1228. <https://doi.org/10.1016/j.apenergy.2018.05.055>.
- Li, Y., Zang, R., Liu, G., Chen, C., He, Y., Liu, X., 2013. Comparison of methane production potential, biodegradability, and kinetics of different organic substrates. *Bioresour. Technol.* 149, 565–569. <https://doi.org/10.1016/j.biortech.2013.09.063>.
- Liu, R., Hao, X., Wei, J., 2016. Function of homoacetogenesis on the heterotrophic methane production with exogenous H₂/CO₂ involved. *Chem. Eng. J.* 284, 1196–1203. <https://doi.org/10.1016/j.cej.2015.09.081>.
- Liu, Y., Gu, M., Yin, Q., Wu, G., 2019. Inhibition mitigation and ecological mechanism of mesophilic methanogenesis triggered by supplement of ferroferric oxide in sulfate containing systems. *Bioresour. Technol.* 288, 121546. <https://doi.org/10.1016/j.biortech.2019.121546>.
- Lizama, A.C., Figueiras, C.C., Pedreguera, A.Z., Espinoza, J.E.R., 2019. Enhancing the performance and stability of the anaerobic digestion of sewage sludge by zero valent iron nanoparticles dosage. *Bioresour. Technol.* 275, 352–359. <https://doi.org/10.1016/j.biortech.2018.12.086>.
- Lohani, S.P., Havukainen, J., 2018. Anaerobic digestion: factors affecting anaerobic digestion process. In: Varjani, S., Gnanounou, E., Gurunathan, B., Pant, D., Zakaria, Z. (Eds.), *Waste Bioremediation, Energy, Environment and Sustainability*. Springer, Singapore. https://doi.org/10.1007/978-981-10-7413-4_18.
- Luo, Y.Q., Hui, D.F., Zhang, D.Q., 2006. Elevated CO₂ stimulates net accumulations of carbon and nitrogen in land ecosystems: a meta-analysis. *Ecology* 87, 53–63. <https://doi.org/10.1890/04-1724>.
- Mayer, F., Gerin, P.A., Noo, A., Foucart, G., Flammang, J., Lemaigre, S., Sinnaeve, G., Dardenne, P., Delfosse, P., 2014. Assessment of factors influencing the biomethane yield of maize silages. *Bioresour. Technol.* 153, 260–268. <https://doi.org/10.1016/j.biortech.2013.11.081>.
- Miranda, N.D., Granel, R., Tuomisto, H.L., McCulloch, M.D., 2016. Meta-analysis of methane yields from anaerobic digestion of dairy cattle manure. *Biomass Bioenergy* 86, 65–75. <https://doi.org/10.1016/j.biombioe.2016.01.012>.
- Miranda, N.D., Tuomisto, H.L., McCulloch, M.D., 2015. Meta-analysis of greenhouse gas emissions from anaerobic digestion processes in dairy farms. *Environ. Sci. Technol.* 49, 5211–5219. <https://doi.org/10.1021/acs.est.5b00018>.
- Morales, C.L., Traveset, A., 2009. A meta-analysis of impacts of alien vs. native plants on pollinator visitation and reproductive success of co-occurring native plants. *Ecol. Lett.* 12, 716–728. <https://doi.org/10.1111/j.1461-0248.2009.01319.x>.
- Nelson, M.C., Morrison, M., Yu, Z., 2011. A meta-analysis of the microbial diversity observed in anaerobic digesters. *Bioresour. Technol.* 102, 3730–3739. <https://doi.org/10.1016/j.biortech.2010.11.119>.
- Ogbuewu, I.P., Okoro, V.M., Mbajioru, C.A., 2019. Meta-analysis of the influence of phytochemical (pepper) supplementation in broiler chicken performance. *Trop. Anim. Health Prod.* 1–14. <https://doi.org/10.1007/s11250-019-02118-3>.
- Puyol, D., Flores-Alsina, X., Segura, Y., Molina, R., Padrino, B., Fierro, J.L.G., Gernaey, K.V., Melero, J.A., Martinez, F., 2018. Exploring the effects of ZVI addition on resource recovery in the anaerobic digestion process. *Chem. Eng. J.* 335, 703–711. <https://doi.org/10.1016/j.cej.2017.11.029>.
- Raposo, F., De la Rubia, M.A., Fernández-Cegri, V., Borja, R., 2011. Anaerobic digestion of solid organic substrates in batch mode: an overview relating to methane yields and experimental procedures. *Renew. Sustain. Energy Rev.* 16, 861–877. <https://doi.org/10.1016/j.rser.2011.09.008>.
- Romero-Güiza, M.S., Vila, J., Mata-Alvarez, J., Chimenos, J.M., Astals, S., 2016. The role of additives on anaerobic digestion: a review. *Renew. Sustain. Energy Rev.* 58, 1486–1499. <https://doi.org/10.1016/j.rser.2015.12.094>.
- Rosenberg, M.S., Rothstein, H.R., Gurevitch, J., 2013. Effect sizes: conventional choices and calculations. In: Koricheva, J., Gurevitch, J., Mengersen, K. (Eds.), *Handbook of Meta-Analysis in Ecology and Evolution*. Princeton University Press, Princeton and Oxford.
- Sabater, S., Bregoli, F., Acuña, V., Barceló, D., Elosegui, A., Ginebreda, A., Marcé, R., Muñoz, I., Sabater-Liesla, L., Ferreira, V., 2018. Effects of human-driven water stress on river ecosystems: a meta-analysis. *Sci. Rep.* 8, 11462. <https://doi.org/10.1038/s41598-018-29807-7>.
- Schmidt, T., Nelles, M., Scholwin, F., Pröter, J., 2014. Trace element supplementation in the biogas production from wheat stillage – optimization of metal dosing. *Bioresour. Technol.* 168, 80–85. <https://doi.org/10.1016/j.biortech.2014.02.124>.
- Shi, X.S., Dong, J.J., Yu, J.H., Yin, H., Hu, S.M., Huang, S.X., Yuan, X.Z., 2017. Effect of

- hydraulic retention time on anaerobic digestion of wheat straw in the semi-continuous continuous stirred-tank reactors. *BioMed Res. Int.* <https://doi.org/10.1155/2017/2457805>.
- Simmonds, M.C., Higgins, J.P.T., 2007. Covariate heterogeneity in meta-analysis: criteria for deciding between meta-regression and individual patient data. *Stat. Med.* 26, 2982–2999. <https://doi.org/10.1002/sim.2768>.
- Strömberg, S., Nistor, M., Liu, J., 2014. Towards eliminating systematic errors caused by the experimental conditions in Biochemical Methane Potential (BMP) tests. *Waste Manag.* 34, 1939–1948. <https://doi.org/10.1016/j.wasman.2014.07.018>.
- Suanon, F., Sun, Q., Li, M., Cai, X., Zhang, Y., Yan, Y., Yu, C.P., 2017. Application of nanoscale zero valent iron and iron powder during sludge anaerobic digestion: impact on methane yield and pharmaceutical and personal care products degradation. *J. Hazard Mater.* 321, 47–53. <https://doi.org/10.1016/j.jhazmat.2016.08.076>.
- Sullivan, G.M., Feinn, R., 2012. Using effect size or why the P value is not enough. *J. Grad. Med. Educ.* 279, 282. <https://doi.org/10.4300/JGME-D-12-00156.1>.
- Ugwu, S.N., Enweremadu, C.C., 2019. Effects of pre-treatments and co-digestion on biogas production from Okra waste. *J. Renew. Sustain. Energy* 11 (1), 013101. <https://doi.org/10.1063/1.5049530>.
- Ugwu, S.N., Enweremadu, C.C., 2019b. Enhancing anaerobic digestion of okra waste with the addition of iron nanocomposite (Ppy/Fe3O4). *Biofuels*. <https://doi.org/10.1080/17597269.2019.1702796>.
- Van Houwelingen, H.C., Arends, L.R., Stijnen, T., 2002. Advanced methods in meta-analysis: multivariate approach and metaregression. *Stat. Med.* 21, 589–624. <https://doi.org/10.1002/sim.1040>.
- Wallace, B.C., Lajeunesse, M.J., Dietz, G., Dahabreh, I.J., Trikalinos, T.A., Schmid, C.H., Gurevitch, J., 2017. Intuitive, open-source software for meta-analysis in ecology and evolutionary biology. *Methods Ecol. Evol.* 8, 941–947. <https://doi.org/10.1111/2041-210X.12708>.
- Wang, Y., Wang, D., Fang, H., 2018. Comparison of enhancement of anaerobic digestion of waste activated sludge through adding nano-zero valent iron and zero valent iron. *RSC Adv.* 8, 27181–27190. <https://doi.org/10.1039/c8ra05369c>.
- Wang, P., Chen, X., Liang, X., Cheng, M., Ren, L., 2018b. Effects of nanoscale zero-valent iron on the performance and the fate of antibiotic resistance genes during thermophilic and mesophilic anaerobic digestion of food waste. *Bioresour. Technol.* <https://doi.org/10.1016/j.biortech.2019.122092>.
- Yang, Y., Guo, J., Hu, Z., 2013. Impact of nano zero valent iron (NZVI) on methanogenic activity and population dynamics in anaerobic digestion. *Water Res.* 47, 6790–6800. <https://doi.org/10.1016/j.watres.2013.09.012>.
- Ye, M., Liu, J., Ma, C., Li, Y.-Y., Zou, L., Qian, G., Xu, Z.P., 2018. Improving the stability and efficiency of anaerobic digestion of food waste using additives: a critical review. *J. Clean. Prod.* 192, 316–326. <https://doi.org/10.1016/j.jclepro.2018.04.244>.
- Yu, B., Zhang, D., Dai, X., Lou, Z., Yuan, H., Zhu, N., 2016. The synthetic effect on volatile fatty acid disinhibition and methane production enhancement by dosing FeCl₃ in a sludge thermophilic anaerobic digestion system. *RSC Adv.* 6 (25), 21090–21098. <https://doi.org/10.1039/c5ra26245c>.
- Zhang, Y., Feng, Y., Yu, Q., Xu, Z., Quan, X., 2014. Enhanced high-solids anaerobic digestion of waste activated sludge by the addition of scrap iron. *Bioresour. Technol.* 159, 297–304. <https://doi.org/10.1016/j.biortech.2014.02.114>.
- Zhang, J., Lu, T., Wang, Z., Wang, Y., Zhong, H., Shen, P., Wei, Y., 2019. Effects of magnetite on anaerobic digestion of swine manure: attention to methane production and fate of antibiotic resistance genes. *Bioresour. Technol.* 291, 121847. <https://doi.org/10.1016/j.biortech.2019.121847>.
- Zhang, Z., Gao, P., Cheng, J., Liu, G., Zhang, X., Feng, Y., 2018. Enhancing anaerobic digestion and methane production of tetracycline wastewater in EGSB reactor with GAC/NZVI mediator. *Water Res.* 136, 54–63. <https://doi.org/10.1016/j.watres.2018.02.025>.
- Zhang, J., Sui, Q., Zhong, H., Meng, X., Wang, X., Wang, Y., Wei, Y., 2018b. Impacts of zero valent iron, natural zeolite and Dnase on the fate of antibiotic resistance genes during thermophilic and mesophilic anaerobic digestion of swine manure. *Bioresour. Technol.* 258, 135–141. <https://doi.org/10.1016/j.biortech.2018.03.005>.
- Zhao, Z., Li, Y., Quan, X., Zhang, Y., 2017. Towards engineering application: potential mechanism for enhancing anaerobic digestion of complex organic waste with different types of conductive materials. *Water Res.* 115, 266–277. <https://doi.org/10.1016/j.watres.2017.02.067>.
- Zhao, Z., Zhang, Y., Li, Y., Quan, X., Zhao, Z., 2018. Comparing the mechanisms of ZVI and Fe3O4 for promoting waste activated sludge digestion. *Water Res.* 144, 126–133. <https://doi.org/10.1016/j.watres.2018.07.028>.
- Zheng, L., Chen, J., Zhao, Cheng, S., Wang, L.-P., Mang, H.-P., Li, Z., 2020. What could China give to and take from other countries in terms of the development of the biogas industry? *Sustainability* 12 (1490), 1–21. <https://doi.org/10.3390/su12041490>.
- Zhu, L., Li, Z., Hiltunen, E., 2017. Theoretical assessment of biomethane production from algal residues after biodiesel production. *WIREs Energy Environ* 2017, e273. <https://doi.org/10.1002/wene.273>.